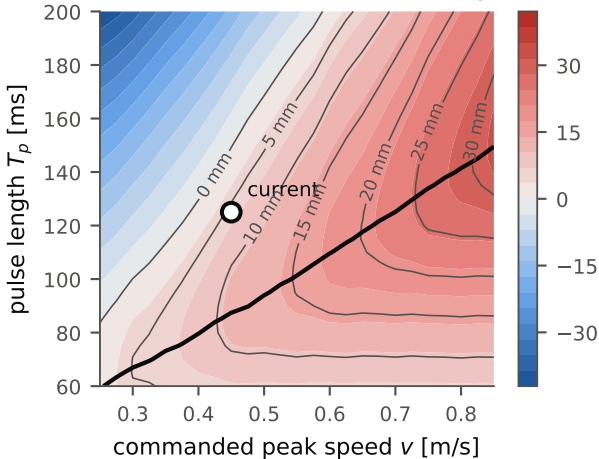


(a) slip, black line: $w = 0$ at re-grip



(b) cost of a pulse that outlasts the apex

